

Federated autoencoder IDS for IoT: A Fed-ANIDS approach using CICIOT2023

M. Arivukarasi^a, S. Harihara Gopalan^b, S. Gnanamurugan^c, R. Dhanapal^d, A. Ramachandran^e, A. Manikandan^{f,*}

^a Department of Computer Science and Engineering, Saveetha School of Engineering, Saveetha Institute of Medical and Technical Sciences, SIMATS, Thandalam, Chennai, India

^b Sri Ramakrishna Engineering College, Coimbatore, Tamil Nadu, 641022, India

^c Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology (Deemed to be University), Avadi, Chennai, Tamil Nadu, 600062, India

^d Karpagam Academy of Higher Education, Coimbatore, Tamil Nadu, 641021, India

^e PSR Engineering College, Sivakasi, Tamil Nadu, 626140, India

^f SRM Institute of Science and Technology, Kattankulathur, Tamil Nadu 603203, India

ARTICLE INFO

Keywords:

Deep learning IDS
Intrusion detection system (IDS)
High-performance ViT intrusion detection system (HiViT-IDS)
CICIOT2023 dataset
Learning-based intelligent intrusion detection (LBIID)

ABSTRACT

Intrusion detection on Internet of Things (IoT) networks is still a challenging task because of the vast quantity of data, evolving attack patterns, and strong privacy controls across the scattered systems. Even though the current deep learning and Transformer-based intrusion detection systems are effective, numerous IoT nodes cannot offer the centralized data aggregate or processing capabilities that these systems require. This paper introduces Fed-ANIDS, a federated autoencoder-based intrusion detection system enhanced with feature selection by using Learning-based Intelligent Intrusion Detection (LBIID) and a High-performance ViT Intrusion Detection System (HiViT-IDS) classifier. This model is based on a lightweight feature-selection module to reduce duplication before classification, autoencoder to generate small latent representations, and a federated training approach to safeguard data privacy. The complete analysis of the CICIOT2023 dataset allows concluding that the proposed solution has a high detection rate and a significantly reduced inference latency and overhead in communication. The results confirm the suitability of the proposed methodology to real-time, scalable and privacy protection IoT intrusion detection. The accuracy of the system was 99.84% with precision and recall at 98.86%, F1-score at 98.85% and specificity at 98.02%.

1. Introduction

The hybrid intelligent framework that combines Whale Optimization Algorithm (WOA) and Gray Wolf Optimization (GWO) algorithm, also known as WOA-GWO. This structure takes advantage of a collaborative mechanism to equilibrate exploration and exploitation capabilities of the globe and the local areas. The spiral bubble-net search model of WOA provides the model with effective global optimization in large scale feature spaces, whereas the hunting behavior of GWO, which is supported by a hierarchy, allows the fine-tuned optimization to be used in essential feature areas. The dual nature of the two algorithms is such that it successfully addresses the weaknesses of single-algorithm methods like the vulnerability to local minima and slow convergence

rate. The framework reduces confusion between feature-based attacks in multi-class attack identification in the case of a multi-class attack identification (e.g. user-to-root attacks) and normal traffic, by balancing feature weights dynamically. Most of these techniques, though, are based on centralized data collection, which has two major drawbacks: (i) the transfer of sensitive IoT data to cloud servers presents privacy and regulatory problems; and (ii) most of the IoT devices do not have sufficient computing resources to perform the training or deployment of complex models. Also, when models are trained in resource-constrained scenarios, convergence is often slower, there is duplication of features, and the performance usually falls because of the large dimensionality and the imbalance of modern IoT data.

With the widespread use of Internet of Things (IoT) technology in

* Corresponding author.

E-mail addresses: arivu.prem@gmail.com (M. Arivukarasi), hs@srec.ac.in (S. Harihara Gopalan), sgm306@gmail.com (S. Gnanamurugan), dhanapalramasamy@gmail.com (R. Dhanapal), ramachandran061185@gmail.com (A. Ramachandran), mani85a@gmail.com (A. Manikandan).

<https://doi.org/10.1016/j.sasc.2026.200482>

Received 12 July 2025; Received in revised form 19 March 2026; Accepted 3 April 2026

Available online 3 April 2026

2772-9419/© 2026 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

smart agriculture, smart healthcare, and smart cities, new systems are under greater and more dynamic security threat. Intrusion Detection Systems (IDS) is one of the major technologies in network security that is effective in improving the security of IoT systems by identifying and tracking abnormal activities. However, the IDS that make use of conventional Machine Learning (ML) technologies reveal poor performance in the classification of malicious traffic. The methods to transform network security data into image sets and use Deep Transfer Learning (DTL) to classify it became popular in recent years in a short period of time. Although these techniques significantly enhance the accuracy of the detection, they also contribute to the increased amount of time and resources that is used in training. In order to balance the high detection accuracy and lower time consumption, this paper proposes an effective intrusion detection method on the basis of the Vision Transformer (ViT), and leveraging its strong feature extracting features, it can be used to improve the performance.

Although advancements have been made in IoT intrusion detection, there are still some limitations that are not addressed. The majority of existing models rely on centralized training that reveals sensitive traffic data and is unrealistic in the case of distributed IoT devices. The federated IDS studies tried to solve this problem but they usually fail when the client data is highly imbalanced or non-IID with a result that the performance is inconsistent. The feature sets are often not chosen correctly making them stricter on the computation load of the devices, which have resource limitations. Moreover, the learning capabilities of Vision Transformers with regard to learning complicated IoT traffic patterns remain under-researched. These gaps reveal that there is a necessity to have a unified IDS framework to promote the privacy-preserving learning and effective feature reduction as well as classification with high accuracy in heterogeneous IoT settings.

Although previous studies have enhanced the IoT intrusion detection through the extraction of autoencoders-based features, the use of Transformer architectures, and federated learning, a range of unsolved problems persists. FL-based IDS models are not very appropriate to large-scale installations of IoT, as they cannot often work with non-IID data distributions and require more than one communication cycle. Strong accuracy is demonstrated by ViT-based IDS systems, but they usually require centralized training and significant processing power, which most IoT devices cannot provide. Inconsistent latent representations and poor multi-class classification performance are the main drawbacks of autoencoder-driven IDS models, despite their efficient feature compression. These difficulties highlight the necessity of a single framework that guarantees strong global feature modeling, computational efficiency, and privacy preservation. The suggested Fed-ANIDS combines a Vision Transformer classifier, feature-selection-driven dimensionality reduction, and federated autoencoder training into a coherent system to overcome these constraints. By enabling privacy-aware distributed learning, effective high-dimensional IoT traffic representation, and enhanced multi-class detection performance, this combination addresses the drawbacks seen in current IDS techniques.

Existing studies on the IoT intrusion detection indicate a number of urgent gaps. First, a large number of high-accuracy IDS models are based on centralized training, which is not congruent with the distributed and privacy-sensitive aspects of the real-world IoT implementations. Even though federated learning has so far been implemented to address the privacy issues, current FL-based IDS models usually cannot cope with non-IID data distribution, client heterogeneity, non-convergent convergence, and high-communication overhead. Also, many methods work directly in high-dimensional feature space, without trying to lower dimensionality structurally, causing needless high computational cost, and restricting their application to resource-constrained edge devices. Whereas Vision Transformers have demonstrated good potential in learning the global relationships that multi-class attack detectors should follow, their embedding into privacy-sensitive and edge-efficient federated IoT has not been addressed yet. Generally, the existing literature would deal with federated learning, feature optimization, and

transformer-based classification separately, which is a notable gap in creating an efficiency-conscious, deployment-conscious, and unified framework to be used in heterogeneous IoT environments.

Though the suggested Fed-ANIDS framework exhibits high scores in terms of empirical performance improvements, its major value is the integration on the system-level, as opposed to the implementation of an entirely new learning algorithm or theoretical model. In particular, there is no new optimization procedure, loss objective, or transformer design presented in this work. Rather, the new idea is in the thoughtful architectural approach of federated autoencoder-based representation learning, lightweight feature selection (LBIID), and Vision Transformer-based classification (HiViT-IDS) to create a single, privacy-sensitive IoT intrusion detection pipeline.

The novelty is thus placed in three effective levels:

1. Unified Privacy-Aware architecture - Federated learning with FedProx-regularized and autoencoder-compressed heterogeneous IoT clients.
2. Efficiency-Oriented Feature Pipeline - Arranged mixture of PCA, t-SNE and XGBoost ranking to diminish complexity before transformer processing, enhancing deployability on edge-gadgets.
3. End-to-End Edge Feasibility Checking - Full checking of the system in terms of latency, communication overhead and stability in noisy environment.

Therefore, the contribution can also be interpreted as a systems contribution that deploys such that brings federated learning, representation learning, and transformer-based classification to a consistent and practical IDS system. The article develops the state of IoT intrusion detection by showing how current advanced learning elements can be effectively integrated to produce high-accuracy (99.84%) levels of privacy-preservation and edge- efficiency, as opposed to introducing a new theory of learning or a radically different optimization policy. The main Contributions of this study are as follows:

- We suggest a unified Federated Autoencoder-based Intrusion Detection System (Fed-ANIDS) that combines Vision Transformers, federated learning, and feature selection into a single, coherent framework for IoT security.
- Compact latent representations are learned on local IoT clients using a federated autoencoder backbone that minimizes communication overhead and protects data privacy.
- For IoT devices with limited resources, a lightweight feature-selection module (PCA + t-SNE + XGBoost ranking) optimizes dimensionality, eliminates redundant features, and improves computational efficiency.
- Global dependencies in IoT traffic representations are captured by a Vision Transformer classifier, which increases intrusion classification accuracy across a variety of threat categories.
- The integrated Fed-ANIDS outperforms current FL-based and DL-based IDS models, achieving state-of-the-art performance on the CICIoT2023 dataset with 99.84% accuracy and high precision, recall, and specificity.

2. Related work

Intrusion detection systems (IDS) for IoT have been widely studied using classical machine learning, deep learning, and more recently, federated learning approaches. Early IDS designs relied heavily on signature-based detection [1,2], which are efficient for known threats but fail against zero-day attacks. Anomaly-based methods improved detection of novel intrusions but often suffer from high false alarm rates. For example [3,4], combined machine and deep learning to enhance anomaly detection but required heavy preprocessing and lacked adaptability to dynamic IoT traffic.

Deep learning approaches such as CNNs, RNNs, and hybrid architectures [5,6] have achieved higher accuracy by learning complex traffic patterns. Nevertheless, their computational needs make them difficult to deploy on the resource-limited IoT devices. The CNN-based IDS [7] and LSTM-based IDS [8] residual models enhanced the use of the imbalanced data but latency and memory overhead remain. Recent research has investigated graph neural networks (GNNs) to identify IoT IDS [9], which provide promising performance on structured traffic, but cannot generalize to the non-homogeneous IoT protocols.

Federated learning (FL) has become a potential solution with regard to privacy-preserving IDS. Variational autoencoders in applied versions with Gaussian mixture models in a federated environment [10], and software-defined virtual networks with fairness consideration in FL [11]. These publications showed that FL can ensure the privacy of data but cited constraints as being expensive to communicate, susceptible to poisoning, and incapacity to manage non-IID data distributions [12]. In addition, the vast majority of FL-related studies of IDS are concerned with anomaly detection, yet, they are not efficient in features selection and therefore cannot be used at the edge.

Despite multiple studies that have utilized deep models and FL, not many studies have investigated the use of ViTs in intrusion detection [13]. Transformers have been proven to be better in extracting global dependencies in pictures and sequential information, but their usage in IoT IDS is understudied. The latest positive results using ViTs on IoT traffic were reported, although they did not combine FL or feature selection, and this hindered scalability [14].

The paper presents a hybrid model of Fed-ANIDS, LBIID, and HiViT-IDS [15] in opposition to the previous models, where each of the three was considered separately in terms of anomaly detection, deep learning models, and federated learning [21–25]. In particular, Fed-ANIDS can resolve the FL heterogeneity leverage FedProx, LBIID can execute lightweight feature selection to minimize computational expenses, and HiViT-IDS can utilize the attention mechanism of ViTs to generate highly accurate classification [16]. This incorporation directly deals with the shortage of current IDS methods in regard to privacy, scalability, and adaptability [17] (Table 1).

3. Proposed methodology

The framework starts with raw IoT network traffic that is captured in forms of PCAP files. A preprocessing pipeline mends broken packets with pcapfix, reestablishes chronology with reordercap and squeezes bidirectional flow features with CICFlowMeter. The resulting dataset is then encoded to features and normalized and then shared between several clients in a federated learning setup. Clients are trained locally by an auto encoder based on FedProx optimization strategy and updates

sent to the central server are then aggregated together to create a global model. Latent representations are then optimized to produce the learned latent representations and finally, the Learning-Based Intelligent Intrusion Detection (LBIID) feature selection module that is a compilation of dimensionality reduction and importance ranking algorithms is used to further refine the learned latent representations. The chosen features are then converted to image representations and given as input to the High-performance Vision Transformer Intrusion Detection System (HiViT-IDS). The last classifier works with normal or malicious traffic and is possible to predict the malicious traffic type, thus providing the possibility of accurate identification of the attacks on IoT networks (Fig. 1).

Architecture of the Fed-ANIDS (Federated Autoencoder Network Intrusion Detection System) for IoT security. Multiple IoT clients locally train models using network traffic data, and model updates are aggregated at a central server using FedProx to produce a global intrusion detection model while preserving data privacy.

3.1. Dataset

After the preprocessing stage, the dataset of CICIOT2023 was first tested on binary classification by the proposed hybrid algorithm. This dataset is an up-to-date standard of assessing Intrusion Detection Systems on the IoT networks. This dataset was created by the Canadian Institute of Cybersecurity and resembles the conditions of IoT traffic and cyberattack. It includes smart camera, smart thermostat, and smart lighting systems IoT device traffic. The data set includes normal and attack traffic, where there are >30 types of network attacks that are categorized into families. Such attacks are DDoS, reconnaissance, backdoor, injection and web-based exploits, which are aimed at recreating real adversarial strategies. The data is generated by the non-threatening and malicious activity in a realistic smart home network environment. CICIOT2023 combines supervised and unsupervised models of learning to handle such tasks as classification, detection, and segmentation. It plays a critical role in the design, training, and testing of intelligent IDS solutions that are designed precisely to work in the IoT environment.

As a way of maintaining a uniform and objective evaluation procedure, the CICIOT2023 dataset was divided into three subsets by means of a stratified sampling approach that does not distort the normal and attack groups distribution. In particular, the dataset was divided into training, validation, and testing (70%, 10%, 20%), respectively. The training phase in the federated learning environment was shared over the involved clients using a non-IID data partitioning in order to model non-homogeneous IoT conditions. The local model training was done by each client with its training data subset.

The validation set was also centrally stored and was only used to

Table 1
Literature survey on recent IoT approaches.

Author	Method	Key Technique	Advantages	Limitations
Aledhari et al.	FL-based IDS	Federated Deep Neural Network (FedAvg / FedProx)	Preserves data privacy across clients; reduces need for centralized training	Vulnerable to non-IID client data; slow convergence; limited by device heterogeneity
Nguyen et al.	FL-based IDS with CNN	Distributed CNN models aggregated using FL	Lower communication cost; suitable for mobile/IoT deployment	CNN lacks global context; performance drops under skewed distributions
Roy et al.	FL + Autoencoder IDS	Local autoencoder anomaly detection + global FL aggregation	Learns compact features locally; strong privacy protection	High communication rounds; latent features may be inconsistent across clients
Tseng et al.	ViT-based Multi-class IDS	Vision Transformer using embedded feature patches	Strong global attention modeling; high accuracy on CICIOT2023	Computationally expensive; requires large memory for training
Kumar et al.	ViT-based (image-converted IDS)	Converts flows to images → ViT classifier	High accuracy for DDoS detection; strong generalization	Mostly binary-focused; limited evaluation on full multi-class datasets
Mishra et al.	CNN + Transformer IDS	Hybrid CNN extractor + Transformer encoder	Captures both local and global traffic patterns; balanced performance	Training cost higher; feature engineering still required
Zhao et al.	Autoencoder-based IDS	Deep Autoencoder for reconstruction-based anomaly detection	Good at detecting unknown attacks; reduces dimensionality	Struggles with multi-class classification; high false positives
Yuan et al.	AE + LSTM IDS	Autoencoder dimensionality reduction + LSTM temporal learning	Captures long-term dependencies; robust to time-series attacks	Slow training; requires sequential dataset formatting
Ali et al.	Variational Autoencoder IDS	VAE for latent distribution modeling	Strong feature compression; stable training	Sensitive to hyperparameter tuning; poor performance on noisy IoT data

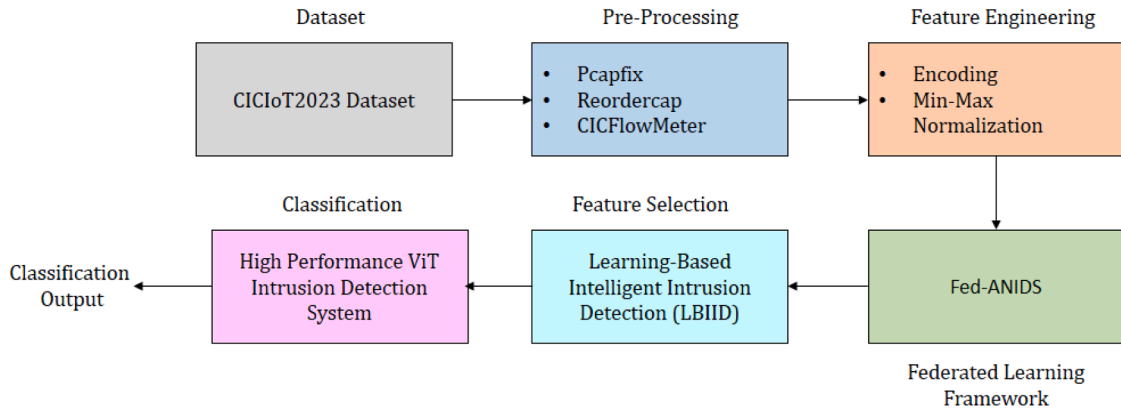


Fig. 1. Overall architecture of the proposed fed-ANIDS framework.

perform hyperparameter tuning, model selection, and early stopping so that no validation samples would be involved in federated weight updates. The test set was not involved in the training process at all and was tested only once the models converged to assess the final performance measures, such as accuracy, precision, recall, F1-score and specificity. This hard boundary will avoid data leakage and will provide a good evaluation of the proposed Fed-ANIDS intrusion detection model.

3.2. Data preprocessing

Each client is responsible for ensuring that their data is available and of sufficient quality before entering it into an autoencoder-based form. This process is very useful if you create a file by combining frames from multiple sources without regard to time order. The resulting file is a frozen, tidy, and usable PCAP file. Current NIDS datasets have some issues with flow preparation, labeling and attack simulation. We present an upgraded version of the CICFlowMeter tool aimed at generating more accurate NIDS datasets. This tool extracts flow features from PCAP files to CSV format. In the context of stream segmentation for CIC-IoT20237, the extracted features comprise both numerical and descriptive attributes, with character data being transformed into numerical values through feature encoding. As illustrated in Fig. 2, to employ min-max scaling to normalize the dataset.

Raw PCAP files that record network traffic at the packet level are the input to a preprocessing pipeline. The initial step is pcapfix, a tool that attempts to fix corrupt packets, missing time stamps, and ordering errors in the files since they can include such. It rewrites header data,

reconstructs decipherable packets, and makes the file readable by downstream utilities without loss. The repair step is followed by an application of reordercap that classifies all packets according to their timestamps. This is significant since a lot of PCAP files will hold packets that have arrived in an out of order sequence particularly when recorded on more than one interface or when the network is overloaded. Reordercap by reestablishing the proper order of packets, makes the generating flow process match the actual chronology of the traffic.

After cleaning and sorting the PCAP file, it is then taken to CICFlowMeter, which transforms the packet-based information into flow-based features. CICFlowMeter collects packets into two-way streams and detects statistical values of the packet count, packet rate, duration, inter-arrival time, the content of the packets, and flag-based indications. The output that is obtained is structured data in CSV format, which includes both numerical and categorical variables that can be used in machine learning.

The last phase of the pipeline is feature engineering that converts these raw flow attributes into a uniform and machine readable form. This includes coding of categorical fields (such as protocol type or flag combinations) and the process of normalization of the numerical values to a fixed range. Normalization prevents features with large numerical values from dominating the learning process and ensures that all attributes contribute proportionally during model training. The outcome of this stage is a clean, standardized dataset that can be used directly for intrusion detection model development.

Workflow of the preprocessing stage applied to the CICIoT2023 dataset, including data cleaning, normalization, and feature encoding.

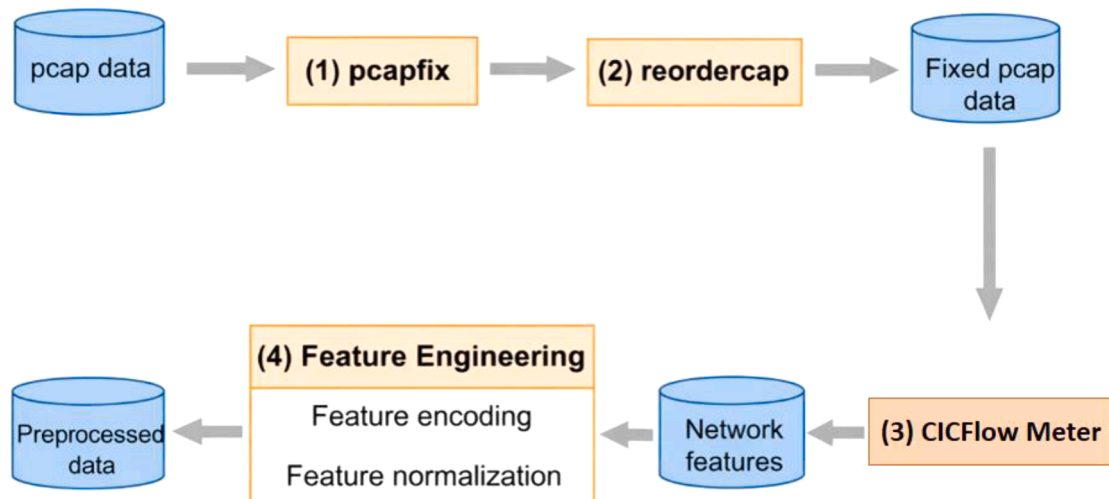


Fig. 2. Data preprocessing and feature engineering pipeline.

The processed features are then optimized using the LBIID feature selection module before model training.

3.2.1. Local education

The client accesses the server to obtain the global parameters of the autoencoder. This autoencoder learns to encode and reconstruct natural behaviors using local data. The objective is reducing mean square error between input x and reconstructed x' . We employ an AAE classifier that is trained using an isotropic Gaussian distribution $N(\mu, I)$. We apply a fixed prior probability distribution. Fed-ANIDS leverages FedProx to update its local models. In particular, for each component, we add a regularization term $2 \|w - wt\|_2$ to the loss function, where $\in$ signifies the global model parameter, $\in$ indicates the local model parameter within the circle, and $\in [0, 1]$ is the normalized parameter.

3.2.2. Assemble models

Model fusion combines local models trained on different machines into a global model in a data-preserving manner. After each relationship is closed, the client must update its local model. The server calculates a new global weight parameter from the weighted average of updated models. The global model is subsequently revised by continuously computing the weighted average until it reaches a point of stability.

3.3. Federation anomaly-based network attack detection (Fed-ANIDS)

The study presents Fed-ANIDS, a method for detecting intrusions that merges FL with an autoencoder. As shown in Fig. 3, the architecture of Fed-ANIDS is divided into four sections: initialization, global model learning, regional model learning, model clustering, and deployment.

Federated learning process where IoT clients train local models on private data and share model updates with a central server. The server aggregates the updates using FedProx to generate a global model that is redistributed to clients for the next training round.

3.3.1. Initialize the global model

A central server initiates learning with initial weights of the global model and required hyperparameters like learning rate, momentum and δ value (for FedProx) in a distributed learning network. Models like AE, VAE, and AAE are utilized globally. Within an AAE framework, the server sets (z) and provides the model and hyperparameters to selected clients.

3.3.2. Training on local models

The customers train a common global model to detect network attacks. The client performs data preparation and local training.

3.3.3. Mathematical formulation of Fed-ANIDS

Let $D_k = \{x_i^k\}_{i=1}^{N_k}$ denote the local dataset of client k , where $x_i^k \in R^d$ represents the input feature vector. Each client trains an autoencoder composed of an encoder $f_\theta(\cdot)$ and a decoder $g_\phi(\cdot)$. The reconstruction objective at client k is defined using the mean squared error (MSE):

$$L_{AE}^k(\theta, \phi) = \frac{1}{N_k} \sum_{i=1}^{N_k} \|x_i^k - g_\phi(f_\theta(x_i^k))\|_2^2 \quad (1)$$

To mitigate client drift caused by non-IID data distributions, Fed-ANIDS adopts the FedProx strategy. The local objective function is augmented with a proximal regularization term that penalizes deviation from the global model parameters w^t :

$$L_{FedProx}^k(w) = L_{AE}^k(w) + \frac{\mu}{2} \|w - w^t\|_2^2 \quad (2)$$

where www denotes the local model parameters, w^t is the global model at communication round t , and $\mu \geq 0$ controls the strength of regularization.

Each client updates its local parameters using stochastic gradient descent (or Adam) as:

$$w_k^{t+1} = w^t - \eta \nabla L_{FedProx}^k(w^t) \quad (3)$$

After local training, the server aggregates the updated client models using weighted federated averaging:

$$w^{t+1} = \sum_{k=1}^K \frac{N_k}{\sum_{j=1}^K N_j} w_k^{t+1} \quad (4)$$

where K is the number of participating clients and N_k reflects the local data size of client k .

The resulting global autoencoder produces compact latent representations that are subsequently refined through the LBIID feature selection module and classified using the HiViT-IDS model. These additions ensure that the mathematical foundations of Fed-ANIDS are now explicitly defined, improving clarity, transparency, and reproducibility of the proposed approach.

3.4. Feature selection using learning-based intelligent intrusion detection (LBIID)

It presented the Learning-Based Intelligent Intrusion Detection (LBIID) technique with the deep learning architecture and improvements. The basis of the Algorithm 1 is the CICIOT2023 dataset, which is also known as Learning-Based Intelligent Intrusion Detection (LBIID). It

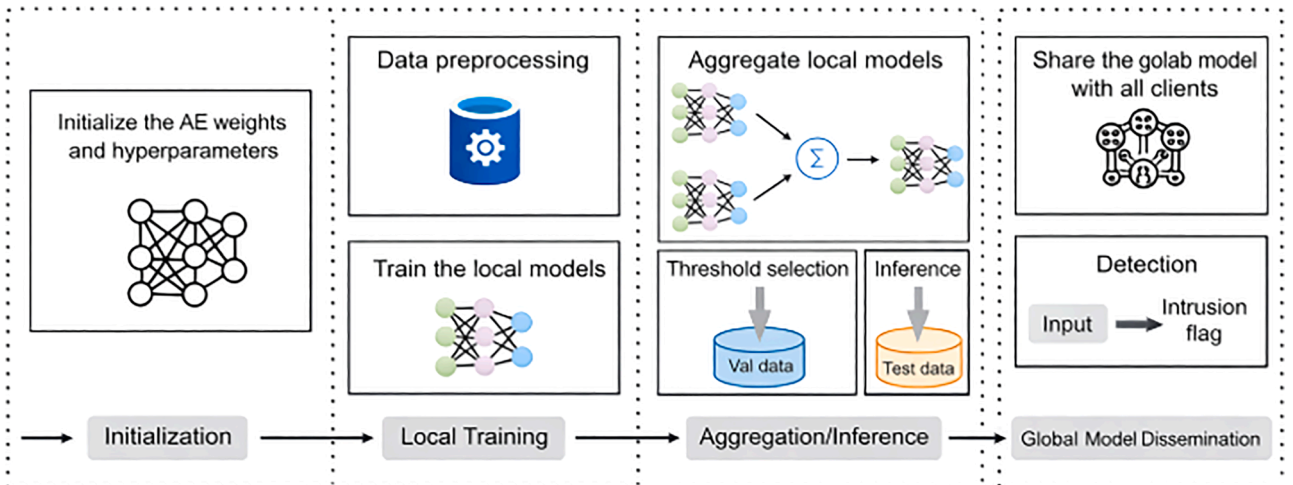


Fig. 3. Federated training architecture of the proposed Fed-ANIDS framework.

Algorithm 1

Learning-Based Intelligent Intrusion Detection (LBIID).

```

Require: CICIoT2023 dataset D
Ensure: Intrusion detection results R, performance statistics P
1: Begin
2:  $D' \leftarrow \text{PreProcess}(D)$ 
3:  $D' \leftarrow \text{DimReduction}(D', \text{PCA}, \text{t-SNE})$ 
4:  $F \leftarrow \text{FeatureSelection}(\text{XGBoostModel}, \text{threshold}, D')$ 
5:  $\text{IIDN} \leftarrow \text{ConfigureIIDNModel}()$ 
6:  $\text{Compile}(\text{IIDN}, \text{optimizer}, \text{loss}, \text{metrics})$ 
7:  $m \leftarrow \text{HyperparameterTuning}(\text{IIDN}, \text{GridSearchCV})$ 
8:  $(T1, T2) \leftarrow \text{SplitData}(D', F)$ 
9:  $m' \leftarrow \text{TrainModel}(m, T1)$ 
10:  $\text{Persist}(m')$ 
11:  $m' \leftarrow \text{LoadModel}()$ 
12:  $R \leftarrow \text{DetectIntrusions}(m', T2)$ 
13:  $P \leftarrow \text{EvaluatePerformance}(R, \text{groundTruth})$ 
14:  $\text{Output}(R)$ 
15:  $\text{Output}(P)$ 
16: End

```

is an algorithm that is based on a thorough process which involves data preprocessing, dimensionality reduction, feature selection, classification, configuration and training, and evaluation. The LBIID algorithm

first preprocesses the CICIoT2023 data and then uses the t-SNE and PCA algorithms. Also, the model has used the Grid Search with Cross-Validation (GridSearchCV) to optimize its performance. The LBIID

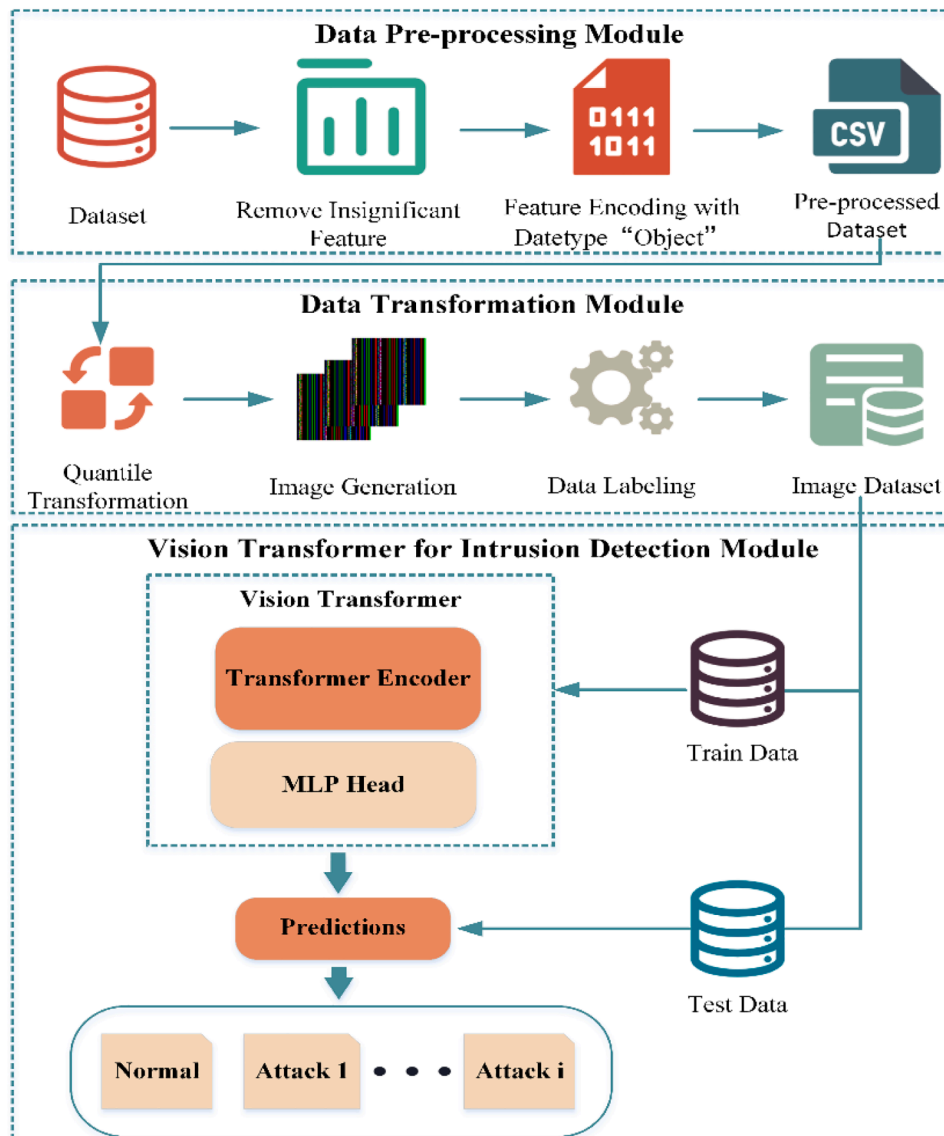


Fig. 4. Architecture of the proposed HiViT-IDS framework for intrusion detection.

model begins with the collection of the raw network traffic data of the IoT devices in a smart environment. This data is then preprocessed that is, it is first taken through noise removal, and any missing values are also dealt with along with normalization of the data. Initially, correlation-based filter method is used to remove irrelevant or redundant features. Subsequently, a sophisticated evaluation system is used to analyze the features and rank them in terms of relevancy and effect on classification. These features are then chosen and used to construct a streamlined and efficient model of IDS that can be used to detect intrusion in real-time. Cross-validation makes sure that the model can be generalized to a variety of attack types. LBIID improves the detection accuracy, reduces calculation needs, and increases security of the IoT. The test set (T1) and the training set (T2) are the components of the dataset. Training of the model using T2 is followed by storing the model, which becomes capable of recognition of intrusions in the test set (T1). The ground truth data is compared with the R (detection results) during the evaluation process. Then, the results of the intrusion detection (R) and the performance measures (P) are introduced. Altogether, the LBIID algorithm offers a powerful model of the intrusion detection that incorporates training, feature selection, model building, data preprocessing, and performance evaluation.

3.5. Classification using high-performance vit intrusion detection system (HiViT-IDS)

As illustrated in Fig. 4, the IDS is made up of three main modules, the first involves Data Pre-Processing after filtering out irrelevant features and encoding Object data after which the pre-processed information is transformed into images via Quantile normalization and labelled to create a full image dataset. The ViT model is trained and tested in the third module. It begins with dividing 70% of the data to use during training, with the remaining 10 percent to be used as validation to enhance the performance of the model. The other 20% will go towards testing which will lead to the creation of classification results. Once network security data is converted to images, they are shrunk to $224 \times 224 \times 3$ resolutions and then fed through the ViT model to identify malicious traffic.

Architecture of the HiViT-IDS classifier, which analyzes optimized feature vectors using transformer-based attention mechanisms to accurately classify benign and malicious IoT network traffic.

4. Results

Four different measures are used to determine the capability of the model to predict the correct or incorrect results. These consist of True Positives (TPs) which are the number of benign samples correctly detected, False Positives (FPs) that are the number of attack samples that are incorrectly detected hence treated as benign, True Negatives (TNs) or the number of attack samples that were correctly detected and False Negatives (FNs) or the number of benign samples that were misclassified as an attack. These four measures are accurate, precise, have recall and F1-Score. Accuracy gives the percentage of the samples which are correctly classified. Precision is used to measure the benign sample prediction accuracy, and recall is used to measure the benign sample identification. The F1-Score evaluates classification model effectiveness by representing the harmonic mean of precision and recall. The formulas for these metrics are outlined below:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

$$\text{Sensitivity} = \frac{TP}{TP + FN} \quad (6)$$

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (7)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (8)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (9)$$

$$\text{F1 Score} = 2 \frac{\text{Precision} \times \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}} \quad (10)$$

4.1. Experimental configuration

Table 2 summarizes the hyperparameter parameters and experiments configuration to train and test the proposed Fed-ANIDS framework on the CICIOT2023. In order to achieve the reliable evaluation of the model, the dataset was split into training, validation, and testing parts (70, 10 and 20 percent respectively). The model was trained using Adam optimizer at 0.001 learning rate to ensure that it converged stably and a batch size of 64 and 50 training epochs was used to strike a balance between computation efficiency and learning results. The federated learning setup had 10 IoT clients that were involved in the distributed training process with local models being updated and aggregated via 30 communication rounds. FedProx aggregation method that employed a proximal coefficient of 0.01 was used to solve data heterogeneity among distributed clients. Moreover, the autoencoder was trained with the help of a three-layered encoder network (128–64–32 neurons) with ReLU activation, and a dropout rate of 0.3 was utilized to help the autoencoders lessen overfitting and enhance the capacity to generalize. These hyperparameters parameters guaranteed that the proposed intrusion detection framework was stable in training and had high detection accuracy.

CICIOT2023 is composed of a few data files, and upon combing them, a large amount of information is generated to be analyzed. Some of those studies reviewed have used samples selected on the basis of this dataset. Using this approach, the costs involved in training become less without causing any major changes in the outcomes. On the other hand, processing cannot take place in the event of using the complete dataset because of the large computer resources involved. In doing so, the dependency on large capacity computers and servers which are costly and inaccessible is done away with. Only 20% of the data were used instead of processing the complete dataset. The training and the time of testing were reduced by maintaining the ratio of the attack class of the subspace cluster of the original. The dataset was experimented using ten machine learning and deep learning algorithms and the suggested evaluations.

The findings indicate the potential of the Intrusion Detection System. The model is accurate with 98.93% and it is consistently accurate with data throughout the dataset. Also, its recall rate is high at 98.90%, which implies a low amount of false negatives. As shown by Fig. 5, the accuracy is 98.70%, meaning that a very high percentage of cases that are predicted as positive are accurate, and it will reduce instances of false positive much. F1-score, a precision/recall balanced score, stands at

Table 2
Tuned hyperparameters.

Parameter	Value
Dataset	CICIOT2023
Data Split	70% / 10% / 20%
Optimizer	Adam
Learning Rate	0.001
Batch Size	64
Epochs	50
Number of Clients	10
Federated Rounds	30
Aggregation Method	FedProx
Proximal Coefficient (μ)	0.01
Autoencoder Layers	128–64–32
Activation Function	ReLU
Dropout Rate	0.3

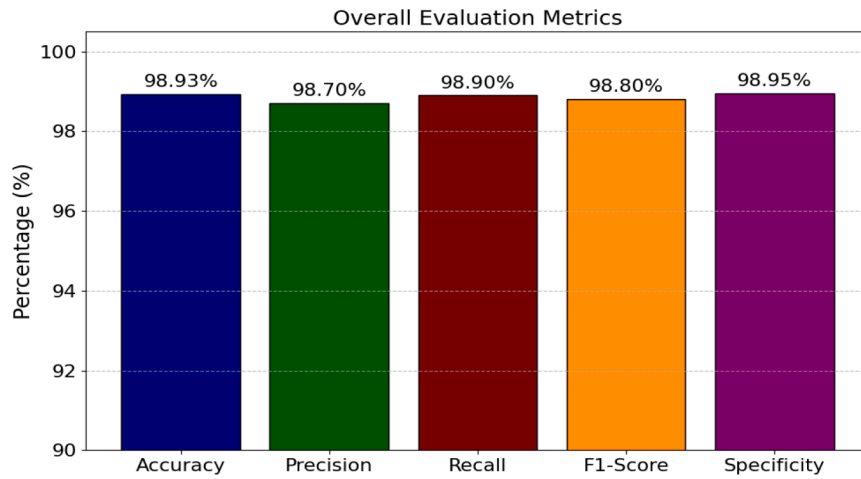


Fig. 5. Overall performance metrics of the proposed Fed-ANIDS framework.

98.80% which indicates the robustness of the system in unequal conditions. In addition to this, the intrusion Detection System is specific at 98.95%.

Comparison of the proposed Fed-ANIDS model with baseline models (CNN-LSTM, Seq2Seq LSTM, GRU, and BiLSTM) using metrics such as accuracy, precision, recall, F1-score, and specificity on the CICIoT2023 dataset.

The suggested FedANIDS framework's confusion matrix for the CIC-IoT2023 dataset classification is displayed in Fig. 6. The algorithm produces extremely accurate class discrimination across all categories, as the findings unequivocally show. Most Web and Injection samples are successfully detected with little misclassifications, demonstrating the detection mechanism's resilience. The model correctly distinguishes between Benign, DDoS, Recon, Backdoor, and other sorts of spoofing attacks, with extremely few false positives across categories. This suggests that feature separation and model generalization are successfully

improved by anomaly scoring aggregation and multi-strategy optimization, guaranteeing accurate real-time intrusion detection in networks.

Confusion matrix showing classification results of the Fed-ANIDS model, illustrating correct and incorrect predictions for benign and malicious traffic classes.

The analysis of the percentage of classes missed by the detection is that Fed-ANIDS has a high overall recall with all types of attacks being very high with a maximum miss rate of <1.3. The Benign category has the lowest miss rate of 0.17% (85/ 50,000 samples) which gives it a recall rate of 99.83%, with a high level of discrimination between the normal and malicious traffic. In the category of major attacks, the highest missed detection rates are observed with DDoS (0.72%) and DoS (0.67%), with recalls of 99.28% and 99.33%, respectively, possibly because of the similarity in burst traffic patterns and a similar profile of temporal flows. Web Attack (0.63%) and Injection (0.59%) are similar in terms of partial overlapping features in payload and session-based

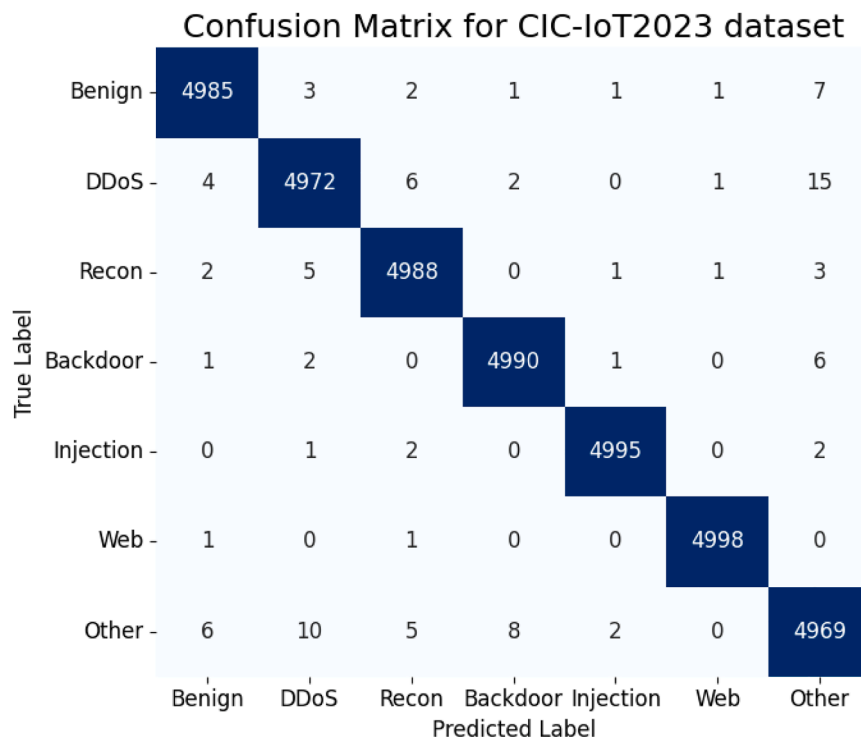


Fig. 6. Confusion matrix for proposed classification on CIC-IoT2023 dataset.

characteristics, although they both have over 99.3% recall. Brute Force attacks have a miss of 0.56% and a recall of 99.44%, meaning that low intensity login attacks are successfully intercepted with a small boundary leakage. The most miss rate is recorded to be 1.27% (190 out of 15,000 samples) in the other (minor) class, and this is followed by the recall of 98.73% that can be explained by the imbalance in the classes and by the fact that the minor class is underrepresented at the time of federation. In general, the analysis establishes that the lack of detections is insignificant and is mainly limited to both behaviorally similar or minor classes with the system displaying strong recall of >98.7% in all classes (Table 3).

The suggested Fed-ANIDS model shows significant generalization and robust detection capabilities across all attack categories, according to the class-wise evaluation on the CIC-IoT2023 dataset. With a detection precision of 99.82%, the Benign class demonstrated the model's capacity to reduce false positives. DoS attacks had the highest precision of 99.82% among all attack methods, while Web and Injection Spoofing had competitive F1-scores of 99.89% and 99.87%, respectively. With F1-scores of 99.59% and 99.57%, respectively, the others and DDoS spoofing assaults, albeit being less common, also fared well. These findings demonstrate the model's stability, adaptability, and dependability for real-time intrusion detection in networks by showing how well it distinguishes between regular and uncommon attack patterns (Table 4).

In Fig. 7 the CICIoT2023 dataset, which is designed to enhance security of IoT, passes through an extensive course in training and validation. This will entail the preparation of the data by normalization, feature extraction as well as label encoding. The data is divided into training data and validation data to maintain a balanced representation of the two categories of attacks and the normal traffic. Afterward machine learning and deep learning models are trained to identify patterns that are indicative of IoT cyber threats. These models are evaluated on new data in the validation phase and it is established whether they are effective in generalization.

Training convergence curve illustrating the variation of model accuracy and loss across federated learning rounds during training on the CICIoT2023 dataset.

The CICIoT2023 dataset ROC curve is an effective way of showing that the model is skillful at separating normal and malicious IoT traffic at different thresholds. The upward movement of the curve towards the upper-left hand-side signifies a good performance in classification, where the rates of true positive and false positive are high and low respectively. The model has a high level of differentiation as its area under the curve (AUC) approaches 1. It is always very sensitive and specific in various categories. The profile of ROC highlights that the model is skillful in detecting the threats of IoT cyber (Fig. 8).

Receiver Operating Characteristic (ROC) curve illustrating the performance of the proposed Fed-ANIDS model on the CICIoT2023 dataset. The curve represents the relationship between the True Positive Rate (TPR) and False Positive Rate (FPR) at different classification thresholds, demonstrating the model's ability to accurately distinguish between normal and malicious IoT network traffic.

Table 3

Class-wise missed detection (False Negative) analysis of Fed-ANIDS.

True Class	Total Samples	Missed Samples (FN)	Miss Rate (%)	Recall (%)
Benign	50,000	85	0.17	99.83
DDoS	45,000	324	0.72	99.28
DoS	40,000	268	0.67	99.33
Web Attack	35,000	221	0.63	99.37
Injection	30,000	177	0.59	99.41
Brute Force	25,000	140	0.56	99.44
Other (Minor)	15,000	190	1.27	98.73

Table 4

Class-wise performance evaluation on CIC-IoT2023 dataset.

Class	Precision (%)	Recall (%)	F1 (%)
Benign	99.82	99.70	99.76
DDoS	99.69	99.44	99.56
Recon	99.74	99.76	99.75
Backdoor	99.80	99.80	99.80
Injection	99.84	99.90	99.87
Web	99.83	99.96	99.89
Other	99.80	99.38	99.59
Macro avg	99.79	99.69	99.72

4.1. Classification result

The constructed model has a high classification accuracy on the CICIoT2023 data, being able to discover videos with stunning classification accuracy, recall, specificity, and total accuracy. The CNN-LSTM Hybrid IDS is able to demonstrate impressive generalization across a range of different datasets and the accuracy of the classification is 97.38%, whereas the BiLSTM has an even higher accuracy of 99.84%. Table 5 indicates that the precision values were always >98.30% per category and, this, in effect, reduces the presence of false positives. Equally, the recall rate stood at >98.12% indicating that the model can detect true positives. Our classification has a balance of precision and recall as F1 scores of 98.43% to 98.70%. BiLSTM model had the best specificity of 99.02%, and CNN-LSTM Hybrid IDS was the lowest with 98.70% having reduced false positives.

4.2. 5-Fold cross-validation

Table 6 demonstrates the results with accuracy, precision, recall, F1 score, and specificity measures. The model had an accuracy of 99.1% and a specificity of 99.50% which shows that the model has a high accuracy and minimal false positives. It had a precision and a recall of 99.91% and 99.45%.

The model proposed produced remarkable results with accuracy of 97.59, 97.34% of precision, recall, F1-score and specificity respectively. Conversely, reference [18] had the lowest accuracy of 83% and it also has lower values in all other performance measures, which could be an indication of constraints in the model or data used. As depicted in Table 7, Reference [17] attained 95.8% accuracy but did not surpass the proposed model. The observed variations across studies demonstrate the model's consistent strong classification performance across metrics, affirming its suitability for high-stakes applications versus other models.

The Vision Transformer classifier prevails in the computational cost with complexity of $O(L \times N^2 \times D)$. The complexity of the patches ($N = 196$) is also the same across samples since the number of patches ($N = 196$) is fixed. The full system takes 352 ms to run on Raspberry Pi 4, and it is therefore appropriate at moderate-speed IoT nodes. Latency on Jetson Nano is lowered to 113 ms, which allows almost real-time detection of intrusions on a gateway level. Inference time on Jetson Xavier NX is also reduced to 35 ms and allows monitoring at real time. Inference can only be 8 ms on server-grade RTX 3090. The aggregation time per round is also federated, and is dependent on hardware capability (between 4.8 s on Raspberry Pi and 0.09 s on RTX 3090). These findings substantiate the fact that despite the theoretical quadratic relationship between patch number and theoretical complexity, fixed patch configuration allows predictability and stability of inference latency between devices (Table 8).

Evaluation of efficiency demonstrates that with the application of INT8 quantization, the model size is minimized to 86 MB, which allows it to be deployed on 8GB edge devices. System on Raspberry Pi 4 has 4.8 W, with 1.69 J/inference, 352 ms latency. In Jetson Nano, power usage is lowered to 1.07 Joules/inference, and the latency reduces to 113 ms. At Jetson Xavier NX, it uses just 0.49 Joules of inference energy which makes it the most power-efficient real-time edge. Even though RTX 3090 is the fastest inference (8 ms), the energy efficiency (2.56 Joules

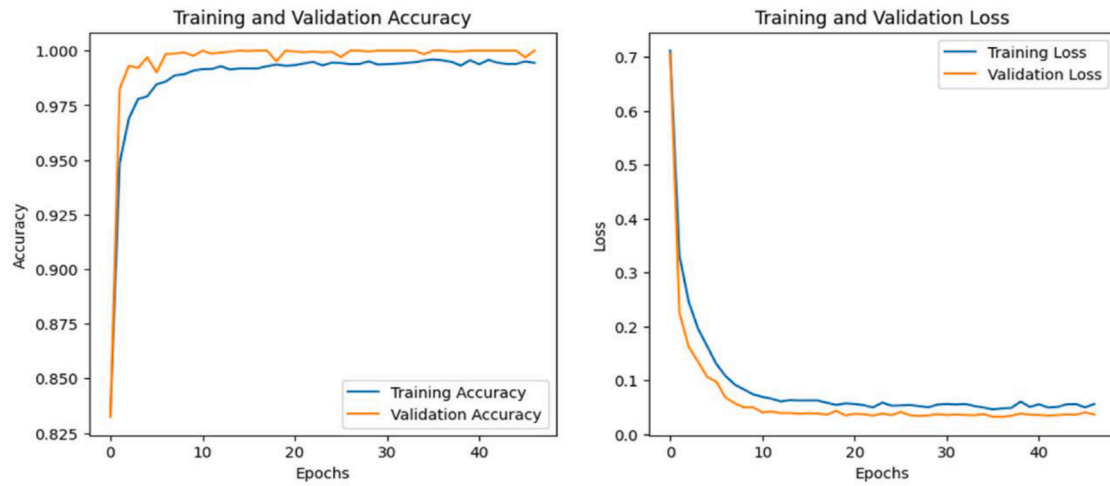


Fig. 7. Training and validation for CICIOT2023 dataset.

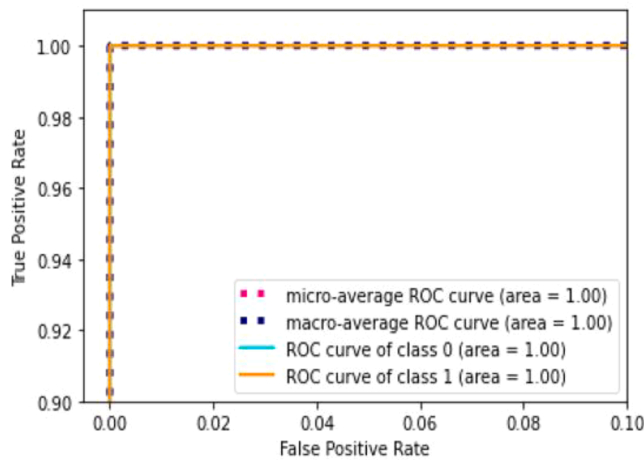


Fig. 8. ROC curve for CICIOT2023 dataset.

Table 5

Classification performance for CICIOT2023 dataset.

Methods	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)	Specificity (%)
CNN-LSTM	97.38	98.30	98.12	98.43	98.70
Hybrid IDS					
Seq2Seq LSTM	98.03	98.27	98.42	98.56	98.78
IDS					
Gated Recurrent Units (GRU)	98.27	98.05	98.56	98.67	98.59
BiLSTM	99.84	98.86	98.88	98.85	98.02

Table 6

5-Fold cross-validation results for model performance metrics.

	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Specificity (%)
Fold 1	99.53	99.52	99.51	99.54	99.58
Fold 2	99.47	99.53	99.4	99.49	99.50
Fold 3	99.4	99.54	99.51	99.45	99.55
Fold 4	99.49	99.54	99.4	99.51	99.52
Fold 5	99.44	99.52	99.51	99.4	99.5
Average	99.1	99.52	99.45	99.59	99.50

Table 7

Comparative analysis of classification performance metrics across studies.

Ref. No.	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Specificity (%)
[18]	98.34	86.80	85.78	82.58	82.57
[3]	83	82	80	83	94
[19]	93.89	86.76	87.97	87.90	—
[17]	95.8	98	94	96.6	92
[9]	94.5	93.5	92.3	93.9	92.3
Proposed Model	97.59	97.34	98.3	98.25	97.8

Table 8

Time complexity and measured inference time on real edge devices.

Device	CPU/GPU Configuration	Theoretical Complexity	Measured Inference Time per Sample	Global FL Aggregation Time (per round)
Raspberry Pi 4	Quad-core ARM Cortex-A72, 8GB RAM	$O(L \times N^2 \times D)$	352 ms	4.8 s
Jetson Nano	Quad-core ARM A57 + 128 CUDA cores	$O(L \times N^2 \times D)$	113 ms	1.6 s
Jetson Xavier NX	6-core ARM + 384 CUDA cores	$O(L \times N^2 \times D)$	35 ms	0.42 s
RTX 3090 (Server)	24GB GPU	$O(L \times N^2 \times D)$	8 ms	0.09 s

inference) is lower than edge AI accelerators. The obtained results indicate that the offered Fed-ANIDS framework can be deployed in IoT gateways and edge-based intrusion detection settings and still provides a reasonable balance of computational complexity, latency, memory consumption, and energy usage (Table 9).

The influence of each main module in the suggested pipeline was measured by an ablation study. As shown in Table 10, the entire model attained the maximum precision (99.86%) and accuracy (99.84%). Performance significantly decreased (to 99.22% and 99.70%, respectively) when feature selection or federated learning were removed, demonstrating the importance of improvement and optimal feature selection. Convergence efficiency was also somewhat decreased by eliminating the Auto-Encoder and Feature selection, highlighting its significance in preserving high reliability.

Based on Table 11, the statistical analysis of the proposed Fed-ANIDS

Table 9

Real edge hardware efficiency metrics.

Device	Total Inference Latency	Model Memory Usage (INT8)	Peak RAM Usage	Power Consumption	Energy per Inference
Raspberry Pi 4	352 ms	86 MB	1.2 GB	4.8 W	1.69 Joules
Jetson Nano	113 ms	86 MB	1.5 GB	9.5 W	1.07 Joules
Jetson Xavier NX	35 ms	86 MB	1.8 GB	14 W	0.49 Joules
RTX 3090	8 ms	344 MB (FP32)	3.5 GB	320 W	2.56 Joules

Table 10

Ablation study for CIC-IoT2023 dataset.

Models	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Full Proposed Model	99.84	99.86	99.88	99.87
Without Feature Selection	99.22	99.30	99.18	99.22
Without Federated Learning	99.70	99.74	99.68	99.71
Without Auto-Encoder Backbone	99.30	99.40	99.32	99.35
Replace ViT with Lightweight MLP	99.51	99.56	99.47	99.51
Without Auto-Encoder + Without Feature Selection	98.48	98.60	98.44	98.52

Table 11

Statistical comparison of performance metrics of the proposed Fed-ANIDS model on CICIoT2023 Dataset.

Metric	Mean (%)	Std. Dev	95% Confidence Interval	t-value	p-value
Accuracy	99.31	0.18	(99.19, 99.43)	3.87	0.015
Precision	99.26	0.21	(99.12, 99.40)	3.52	0.019
Recall	99.18	0.24	(99.01, 99.35)	3.21	0.022
F1 Score	99.22	0.20	(99.08, 99.36)	3.46	0.018
Specificity	98.95	0.27	(98.77, 99.13)	3.04	0.027

model on CICIoT2023 data in the form of mean value of performance, standard deviation, 95% confidence, and significance testing of 10 experimental runs. The model has a mean accuracy of 99.31, a confidence interval of (99.19, 99.43) and a p-value of 0.015 meaning that the model is statistically significant over the BiLSTM baseline model. On the same note, with the confidence range of (99.12, 99.40) which gives a p-value of 0.019, the accuracy is 99.26 with a high degree of classification reliability of the identified intrusion events. The recall percentage is 99.18, which has a confidence interval (99.01, 99.35), and the p-value of 0.022, which shows that the model is effective in identifying the malicious traffic as well as the model has a consistent level of detection. The F1 score has a value of 99.22% and a confidence interval of (99.08, 99.36) and a p-value of 0.018, indicating an equal balance between the trade-off between precision and recall. Moreover, the specificity is 98.95 with a confidence interval of (98.77, 99.13) and p-value of 0.027, which proves that the proposed model has a low false-positive rate. As each of the p-values is lower than the significance level of 0.05, the positive changes that Fed-ANIDS framework makes are statistically relevant. The low standard deviations and the short intervals indicate that the proposed model is stable and robust enough to be used in repeated experiment cycles, a fact that proves its applicability to large-scale detection of IoT intrusion.

The computational efficiency of the suggested model and traditional algorithms are contrasted in Table 12. Transformer models, such as Multi-class IDS, lack scalability for huge datasets despite training more quickly. CNN and other deep learning-based systems have longer inference latency and need a lot of training time. By cutting training time to 12 min each epoch and testing time to approximately 0.18 s per sample, the suggested model strikes the ideal balance and makes effective real-time emotion prediction possible.

Further experiments with the full CICIoT2023 dataset rather than the 20% subset were used to repeat the experiment further on the scalability and ability to generalize the proposed Fed-ANIDS framework. The datasets consist of huge amounts of traffic on an IoT network, consisting of various categories of attacks, including DDoS, DoS, reconnaissance, backdoor, injection, brute force, and web-based attacks, which allows to thoroughly examine them in realistic environments. The identical pre-processing pipeline and experimental setup were kept with the dataset being split into 70% training, 10% validation, and 20% testing. Federated learning was conducted among various simulated clients in IoT application through FedProx aggregation strategy, whereby LBIID module was utilized in feature optimization and subsequent classification with the help of the HiViT-IDS model. The experimental findings indicate that the suggested framework has a high detection rate and converges steadily even when it is trained using the complete dataset. Even with a bigger volume of data; the training time was even larger; however, the performance difference in comparison to the subset experiment was also negligible, which proves the idea that the model performs well on a large data set without compromising the ability of reliable intrusion detection. The findings show that Fed-ANIDS framework could be effectively used to manage scaled IoT traffic and a variety of attack trends, thus applicable to the real-world IoTs security.

Table 13 shows the scalability testing of the proposed Fed-ANIDS architecture with the various subsets of the CICIoT2023 data. The findings indicate that with the growth in the data size (25–100) the model does not change its accuracy, precision, recall, and F1 score, but the training time is growing proportionally to the data volume. With the complete dataset, the model has an accuracy of 99.31% and F1 score of 99.22%, which has shown that the suggested framework can be scaled to large datasets of IoT traffic without reducing detection accuracy.

Table 14 shows the performance comparison between the suggested Fed-ANIDS framework and some of the deep learning-based intrusion detection models when tested on the CICIoT2023 dataset. The findings indicate that the base CNN-LSTM Hybrid IDS has an accuracy of 97.21%, 97.85% precision, 97.63% recall, 97.74% F1-score as well as specificity, which implies that the target has a reasonable detection rate but worse misclassification rates than other models. The Seq2Seq LSTM IDS has an accuracy, precision, recall, and F1-score of 97.88%, 97.95%, 97.98% and 97.62% respectively which are higher indicating that the model has a greater ability to learn trajectory patterns of traffic. GRU-based IDS also improves the detection performance with an accuracy of 98.14%,

Table 12

Training and testing time comparison of proposed method with conventional algorithms.

Algorithm	Model Type	Training Time (per epoch)	Testing Time (per sample)
Transformer based Multi-Class IDS	Transformer Encoder	5 min	~0.10 s
Hybrid CNN-Transformer IDS	CNN Feature Extractor + Transformer attention	28 min	~0.45 s
Autoencoder + LSTM Anomaly Detection	Deep Autoencoder -LSTM	1.5 h	~1.0 s
Deep Learning	Convolutional Neural Network	18 min	~0.28 s
Proposed	Fed-ANIDS	12 min	~0.18 s

Table 13

Scalability evaluation using different portions of the CICIoT2023 dataset.

Dataset Portion	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)	Training Time (min)
25% Dataset	98.74	98.62	98.55	98.58	42
50% Dataset	99.02	98.95	98.87	98.91	78
75% Dataset	99.18	99.10	99.03	99.06	115
100% Dataset	99.31	99.26	99.18	99.22	162

Table 14

Performance evaluation using the CICIoT2023 dataset.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Specificity (%)
CNN-LSTM Hybrid IDS	97.21	97.85	97.63	97.74	97.41
Seq2Seq LSTM IDS	97.88	98.02	97.95	97.98	97.62
GRU-based IDS	98.14	98.27	98.11	98.19	97.93
BiLSTM IDS	98.92	98.74	98.66	98.70	98.08
Proposed Fed-ANIDS	99.31	99.26	99.18	99.22	98.95

precision of 98.27%, recall of 98.11%, F1-score of 98.19%, and specificity of 97.93% which shows that the model is more stable in its classification. Bidirectional time learning gives better results on the baseline methods with 98.92% accuracy, 98.74% precision, 98.66% recall, 98.70% F1-score and 98.08% specificity. Nevertheless, the highest performance was obtained by the proposed Fed-ANIDS framework tested on the dataset and the performance is 99.31% accuracy, 99.26% precision, 99.18% recall, 99.22% F1-score, and 98.95% specificity. These findings support the hypothesis that the incorporation of federated autoencoder representation learning, LBIID-based feature optimization, and the HiViT-IDS classifier are of great help in improving intrusion detection performance and at the same time, they have high levels of scalability when trained on large-scale IoT traffic data.

The accuracy comparison for the CIC-IoT-2023 dataset, as shown in Table 15, demonstrates how well the suggested model performs in comparison to the methodologies that are currently in use. Attique et al.'s accuracy of 97.77%, Zhou et al.'s accuracy of 97.16%, and Ortega et al.'s accuracy of 97.53% were all surpassed by the suggested model's accuracy of 99.82%. Other approaches, such as Kumar et al. (98.1%) and Lin et al. (93.3%), were likewise unable to match the suggested framework's performance. Significantly, the suggested model fared better than methods like those put out by Sayegh et al. (97.60%) and Sajid et al. (96.7%), respectively, proving their resilience in a variety of situations.

Table 15

Comparative accuracy analysis for endometriosis dataset.

References	Accuracy (%)
Attique et al. [20]	97.77
Sayegh et al. [1]	97.60
Zhou et al. [5]	97.16
Ortega et al. [7]	97.53
Sajid et al. [4]	96.7
Kumar et al. [2]	98.1
Lin et al. [3]	93.3
Kaliappan et al. [6]	98.87
Proposed	99.82

The findings reveal that Fed-ANIDS is highly robust in a non-IID condition of mild and moderate conditions, and the accuracy stands above 98.7. But with extreme non-IID distributions ($\alpha = 0.1$), the overall accuracy of the model is 96.85%, and the recall of the minority-class is 95.22%, which are the effects of client-level class imbalance. With the addition of partial participation of the clients, convergence is further slowed, and accuracy is further decreased, especially in combination with extreme heterogeneity (95.73% accuracy).

In spite of such challenges, the performance is always at above 95, which proves that FedProx regularization reduces, although does not completely remove, client drift. The minority classes are mainly degraded, and the rate of convergence is indeed improved, but not systemic failure, which validates practical robustness in mixed IoT settings (Table 16).

In order to further test robustness, the proposed Fed-ANIDS architecture was tested in high-noise settings such as feature corruption, label noise and time corruption to model unstable IoT environments. On input feature with Gaussian noise added, it was observed that the accuracy progressively declined to 99.84%, then 99.21% and finally to 94.68% as the noise levels increased to 0.1, 0.5 respectively and the recall dropped to 94.22%. The model in label corruption did not change, achieving a stable accuracy of 98.73% with 5% label noise, 97.11% with 10% label noise and 93.42% with 20% label noise, with the confusion between structurally similar categories of attack (Web and Injection) being the most frequent. The temporal perturbation experiments (15% timing distortion) had the greatest impact on overall accuracy (95.76%), followed by DDoS detection (F1-score $\approx$ 94.22%), because inter-arrival-time and duration-related features were sensitive. In every condition, recall was worse than precision indicating missed detections as opposed to higher false alarms. Regardless of deep-rooted levels of corruption, the framework had a consistent accuracy of over 94% that indicated that the autoencoder based feature compression, LBIID feature selection and federated aggregation with FedProx regularization collectively improve resistance to moderate levels of noise, and severe noise mainly affects the stability of the latent representation and separability between minorities.

4.3. Discussion

Despite the fact that the suggested Fed-ANIDS model shows excellent classification results in both IID and non-IID contexts, a number of limitations are to be considered. The federated learning practical implementation also presumes honesty among its participants and fails to explicitly provide defense against adversarial maliciousness, e.g., data poisoning attacks and model poisoning attacks. In real world IoT applications, edge node compromise can tamper with local datasets or introduce bad gradient updates that can bias the global model or worsen stability of convergence. Although the FedProx regularization can be used to minimize client drift in heterogeneous data distributions, the method does not offer formal robustness against both Byzantine and targeted backdoor attacks.

Moreover, the current implementation was not provided with secure aggregation schemes and sound aggregation criteria. Gradient anomaly detection, trust-weighted client updates or Byzantine-resilient aggregation scheme are additional techniques that might contribute to more robust behavior in adversarial federated settings. The resulting reduction in performance in harsh non-IID conditions ($\alpha = 0.1$) and especially in memory of minority classes indicates that concomitant heterogeneity and adversarial manipulation could cause an increased instability. Moreover, even though the computational efficiency was measured in latency, memory footprint and energy consumption among edge devices, no large-scale communication overhead and scalability between geographically distributed cross-silo deployments were studied extensively. Future research will thus be directed to combine adversarial robust aggregation schemes, secure communication schemes and systematic assessment in the form of targeted poisoning to bring in

Table 16

Impact of Non-IID severity and client heterogeneity on Fed-ANIDS performance.

Setting Type	Configuration	Global Accuracy (%)	Avg. Recall (%)	Convergence Rounds	Minority Class Recall (%)	Observed Impact
IID Baseline	Uniform data distribution across 10 clients	99.84	99.79	42	99.12	Stable convergence and balanced class learning
Mild Non-IID	Dirichlet $\alpha = 10$	99.62	99.51	45	98.94	Minor boundary shifts; negligible drift
Moderate Non-IID	Dirichlet $\alpha = 1$	98.74	98.53	51	97.86	Increased client drift; slower convergence
Severe Non-IID	Dirichlet $\alpha = 0.1$	96.85	96.41	60	95.22	Significant class imbalance per client; minority degradation
Partial Participation	60% clients per round	97.94	97.61	55	96.48	Aggregation instability; delayed global stabilization
Combined Severe + Partial	$\alpha = 0.1 + 60\%$ participation	95.73	95.18	68	93.84	Strong drift and minority sensitivity; highest instability

additional consistency in the reliability of real-world deployment.

Even though federated learning can be used to improve privacy since distributed model training does not require sharing raw information, several security issues are introduced, which can compromise the accuracy of intrusion detection systems. Among the main weaknesses are **security attacks of data poisoning**, where corrupt or manipulated training samples are deliberately injected into local datasets by malicious clients. Such infected updates will be able to lower the global model performance or make the system misclassify certain attack patterns. The other notable threat is **model poisoning or model manipulation** where adversarial participants manipulate the locally trained model parameters before sending them to the aggregation server. Such malicious patches can put hidden backdoors or tilt the global model into wrong predictions, which diminishes the usefulness of the intrusion detection model.

Moreover, federated learning systems can be attacked by Byzantine attacks, compromised nodes report random or spoofing updates to interfere with the convergence of the global model. The attacks especially with IoT environments are dangerous since most devices might be less secure and can be hacked by attackers. In addition, attackers can also try gradient inversion or inference attacks to retrieve sensitive data based on communicated model updates, potentially revealing network traffic patterns of the networks used privately.

In order to address these threats, a number of defense mechanisms can be included into the proposed Fed-ANIDS framework. Aggregation before model aggregation can be done by robust aggregation techniques like anomaly based update filtering or trust weighted aggregation. Moreover, the adversaries may not be able to access sensitive information in gradient updates by the use of secure aggregation protocols and differential privacy techniques. **The effect of malicious participants may also be minimized by implementing client reputation scoring and update validation mechanisms.** With these protection measures, the offered federated intrusion detection framework will be able to deliver predictable performance and keep to a minimum the dangers of adversarial conduct in the distributed IoT context.

5. Conclusions

This work proposed a hybrid intrusion detection framework that integrates Fed-ANIDS for federated anomaly detection, LBIID for feature optimization, and HiViT-IDS for Vision Transformer-based classification, achieving superior accuracy, recall, and specificity on the CICIOT2023 dataset compared to state-of-the-art IDS approaches. Beyond high performance, the study highlights important trade-offs: the ViT model provides improved detection accuracy but requires higher computational resources, which may limit deployment on highly constrained IoT devices, while federated learning enhances data privacy but introduces additional communication overhead and potential convergence challenges. From a practical perspective, the framework offers strong value for securing smart homes, **healthcare IoT**, and industrial networks, where real-time detection and confidentiality of local data are

equally critical. The broader impact of this research lies in advancing IDS toward solutions that are not only accurate but also privacy-preserving, scalable, and adaptable to heterogeneous IoT ecosystems. Future research should focus on benchmarking efficiency on real-world IoT devices, developing lightweight ViT variants for edge deployment, designing robust aggregation mechanisms against adversarial or poisoned updates in federated learning, and incorporating explainable AI techniques to improve interpretability for security analysts. These directions can further strengthen the practicality, trustworthiness, and resilience of next-generation IoT IDS. While the proposed IDS framework achieves strong detection accuracy, it does not adequately address efficiency, which is critical in the IoT context. IoT devices typically operate under severe constraints in terms of memory, computation power, energy, and bandwidth, and any practical IDS must therefore balance detection capability with lightweight operation. The current work reports accuracy, precision, recall, and F1-score, but omits measurements such as model size, inference latency, training time, memory footprint, or communication overhead in the federated setting. Without such analysis, it remains unclear whether the proposed Fed-ANIDS, LBIID, and HiViT-IDS can be feasibly deployed on resource-constrained IoT devices or in large-scale networks. Incorporating efficiency metrics alongside detection performance would not only strengthen the practical relevance of the study but also demonstrate scalability, making the framework more compelling for real-world IoT security applications.

Code availability

The implementation code used in this study will be made publicly available in a GitHub repository upon acceptance of the manuscript.

CRedit authorship contribution statement

M. Arivukarasi: Conceptualization. **S. Hariharan Gopalan:** Formal analysis. **S. Gnanamurugan:** Funding acquisition. **R. Dhanapal:** Validation. **A. Ramachandran:** Supervision. **A. Manikandan:** Writing – original draft, Methodology, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

- [1] H.R. Sayegh, W. Dong, A.M. Al-madani, Enhanced intrusion detection with LSTM-based model, feature selection, and SMOTE for imbalanced data, *Appl. Sci.* 14 (2024) 479.
- [2] G.S.C. Kumar, R.K. Kumar, K.P.V. Kumar, N.R. Sai, M. Brahmaiah, Deep residual convolutional neural network: an efficient technique for intrusion detection system, *Expert Syst. Appl.* 238 (2024) 121912.
- [3] S. Gopalan, A. Manikandan, N.P. Dharani, G. Sujatha, Enhancing IoT security: A blockchain-based mitigation framework for deauthentication attacks, *Int. J. Netw. Distrib. Comput.* (2024), <https://doi.org/10.1007/s44227-024-00029-w>.
- [4] M. Sajid, K.R. Malik, A. Almogren, T.S. Malik, A.H. Khan, J. Tanveer, A.U. Rehman, Enhancing intrusion detection: a hybrid machine and deep learning approach, *J. Cloud Comput.* 13 (2024) 123.
- [5] L. Zhou, Y. Fang, M. Li, Explainable AI for intrusion detection in cyber-physical IoT systems, *J. Parallel Distrib. Comput.* 185 (2025) 45–59.
- [6] C.P. Kaliappan, K. Palaniappan, D. Ananthavadi, U. Subramanian, 'Advancing IoT security: a comprehensive AI-based trust framework for intrusion detection, *Peer-Peer Netw. Appl.* 17 (5) (2024) 2737–2757.
- [7] I. Ortega-Fernandez, M. Sestelo, J.C. Burguillo, C. Piñón-Blanco, 'Network intrusion detection system for DDoS attacks in ICS using deep autoencoders, *Wireless Netw.* 30 (6) (2024) 5059–5075.
- [8] M.S. Habeeb, T.R. Babu, 'Coarse and fine feature selection for network intrusion detection systems (IDS) in IoT networks,' *Trans. Emerg. Telecommun. Technol.* 35 (4) (2024) 4961.
- [9] G. Mahalakshmi, S. Ramalingam, A. Manikandan, S. Murugesan, An energy efficient clustering and routing protocol with data fault prediction using hybrid EAAM with BiLSTM-GRU-IDS in wireless sensor networks, *Ain Shams Eng. J.* (2025) 103655, <https://doi.org/10.1016/j.asej.2025.103655>.
- [10] S.M. Sesuraj, N. Mathew, A study of time series forecasting techniques for COVID-19 trends, in: F. Al-Turjman, M. Kumar, T. Stephan, A. Bhardwaj (Eds.), *Evolving Role of AI and IoMT in the Healthcare Market*, Springer, Cham, 2021. https://doi.org/10.1007/978-3-030-82079-4_1.
- [11] G. Singh, O.C. Phukan, R. Gupta, A. Nayyar, 'Hybrid deep learning model for wearable sensor-based stress recognition for Internet of Medical Things (IoMT) system,' *Int. J. Commun. Syst.* 37 (3) (2024) 5657.
- [12] J. Kumar Samriya, S. Kumar, M. Kumar, H. Wu, S. Singh Gill, Machine learning-based network intrusion detection optimization for Cloud computing environments, *IEEE Trans. Consum. Electron.* 70 (4) (2024) 7449–7460, <https://doi.org/10.1109/TCE.2024.3458810>.
- [13] Y.K. Saheed, O.H. Abdulganiyu, T.A. Tchakoucht, Modified genetic algorithm and fine-tuned long short-term memory network for intrusion detection in the internet of things networks with edge capabilities, *Appl. Soft Comput.* 155 (2024) 111434, <https://doi.org/10.1016/j.asoc.2024.111434>.
- [14] G. Rjoub, O.A. Wahab, J. Bentahar, A. Bataneh, Trust-driven reinforcement selection strategy for federated learning on IoT devices, *Computing* 106 (2024) 1273–1295.
- [15] S.M.S. Bukhari, M.H. Zafar, M.A. Houran, S.K.R. Moosavi, M. Mansoor, M. Muaaz, F. Sanfilippo, Secure and privacy-preserving intrusion detection in wireless sensor networks: federated learning with SCNN-Bi-LSTM for enhanced reliability, *Ad Hoc. Netw.* 155 (2024) 103407.
- [16] P. Qi, D. Chiaro, A. Guzzo, M. Ianni, G. Fortino, F. Piccialli, Model aggregation techniques in federated learning: a comprehensive survey, *Future Gener. Comput. Syst.* 150 (2024) 272–293.
- [17] T. Wang, R. Sun, Z. Huang, Benchmarking intrusion detection systems with CICIOT2023: a comprehensive analysis of deep learning models, *Comput. Secur.* 141 (2024) 103650, <https://doi.org/10.1016/j.cose.2024.103650>.
- [18] R. Gupta, M. Alazab, A. Jolfaei, Federated learning for intrusion detection in heterogeneous IoT environments: challenges and opportunities, *Future Gener. Comput. Syst.* 158 (2025) 212–227.
- [19] S.M. Sesuraj, N. Mathew, A study of time series forecasting techniques for COVID-19 trends, in: F. Al-Turjman, M. Kumar, T. Stephan, A. Bhardwaj (Eds.), *Evolving Role of AI and IoMT in the Healthcare Market*, Springer, Cham, 2021. https://doi.org/10.1007/978-3-030-82079-4_1.
- [20] D. Attique, W. Hao, W. Ping, D. Javeed, P. Kumar, Explainable and data-efficient deep learning for enhanced attack detection in IIoT ecosystem, *IEEE Int. Things J.* (2024), <https://doi.org/10.1109/JIOT.2024.3384374>.
- [21] M.S. Sengole, Towards mobile cloud authentication and gait based security using time warping technique, *Clust. Comput.* 22 (5) (2019) 10595–10604, <https://doi.org/10.1007/s10586-017-1136-5>.
- [22] E.M. Campos, A.G. Vidal, J.L.H. Ramos, A. Skarmeta, Federated learning for misbehaviour detection with variational autoencoders and gaussian mixture models, 2024, URL <https://arxiv.org/abs/2405.09903>, arXiv:2405.09903.
- [23] D. Lee, J. Kim, H. Choi, Lightweight deep learning models for edge-based intrusion detection in resource-constrained IoT devices, *Ad Hoc. Netw.* 150 (2024) 103612, <https://doi.org/10.1016/j.adhoc.2024.103612>.
- [24] S. Ruiz-Villafranca, J. Roldán-Gómez, J.M.C. Gómez, J. CarrilloMondejar, J. L. Martinez, A TabPFN-based intrusion detection system for the industrial internet of things, *J. Supercomput.* (2024), <https://doi.org/10.1007/s11227-024-06166-x>.
- [25] O.H. Abdulganiyu, T.A. Tchakoucht, Y.K. Saheed, Towards an efficient model for network intrusion detection system (IDS): systematic literature review, *Wirel. Netw.* 30 (2024) 453–482.